

PARKTOWN BOYS' HIGH SCHOOL



Department of Mathematics

June Examination Paper 2 Answer Book

NAME: Marking Guidelines

CLASS: 12 _____ TEACHER: _____

Question	Mark			Moderator
	H	T	U	
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
TOTAL (out of 150)				
Percentage				

QUESTION 1

Ranking of the top 10 Rugby 7s	Points
South Africa	38
Argentina	34
Australia	30
Fiji	26
Spain	26
New Zealand	26
France	14
Kenya	14
Uruguay	6
Germany	5

	Solution	Marks
1.1	$\bar{x} = 21,9$ ✓	(1)
1.2	$\sigma = 10,89$ ✓	(1)
1.3	$[\bar{x} - \sigma ; \bar{x} + \sigma]$ $[11,01 ; 32,79]$ ∴ 6 teams ✓	(3)
1.4.1	Negatively skewed ✓ (left) Median > Mean ✓	(2)
1.4.2	Median ✓	(1)
		[8]

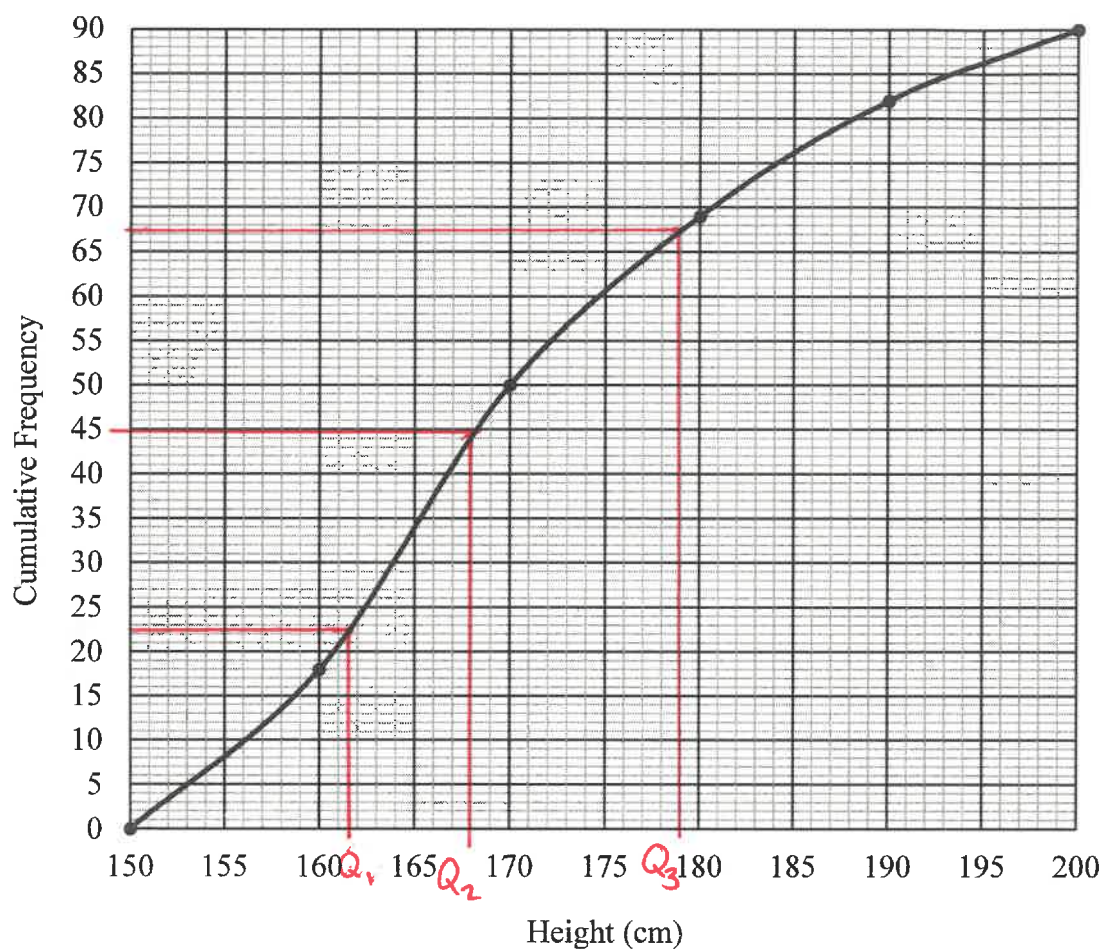
QUESTION 2

Shots on goal by defenders (x)	2	5	7	7	10
Goal difference (y)	-1	0	-1	0	3

	Solution	Marks
2.1	$r = 0,76$ ✓A	(2)
	Moderate positive linear	
	✓ Moderate positive	
2.2	$\hat{y} = a + bx$ ✓m	(3)
	$\hat{y} = -2,44 + 0,43x$	
	✓A ✓B	
2.3	For each additional shot on goal	(1)
	by a defender, the goal difference	
	increases by approximately	
	0,43 goals.	[6]

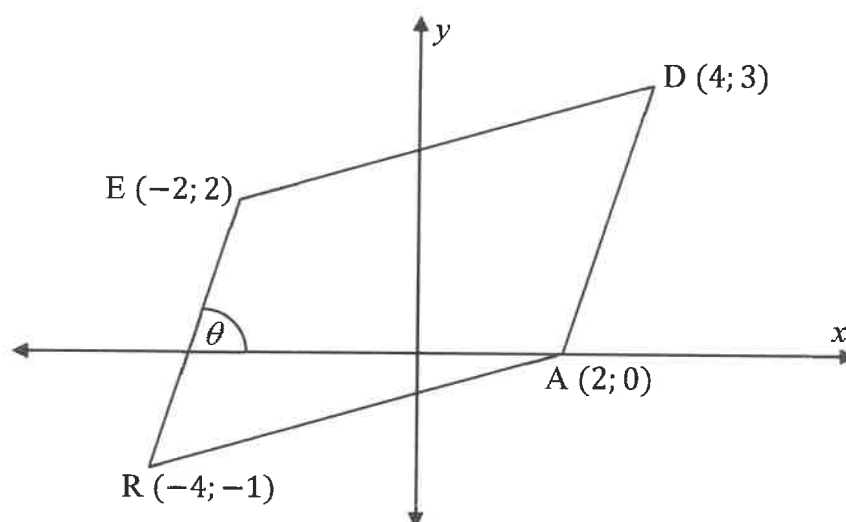
✓ Increases by $\pm 0,43$ goals

QUESTION 3



	Solution	Marks
3.1	90 ✓	(1)
3.2	168cm ✓ ±1cm	(1)
3.3	$Q_1 = 162 \checkmark \pm 1$ $Q_3 = 179 \checkmark \pm 1$ $IQR = 179 - 162$ $= 17\text{cm} \checkmark Q_3 - Q_1$	(3)
3.4	$\frac{21 \checkmark}{90} \times 100 \approx 23\% \checkmark$	(2)
		[7]

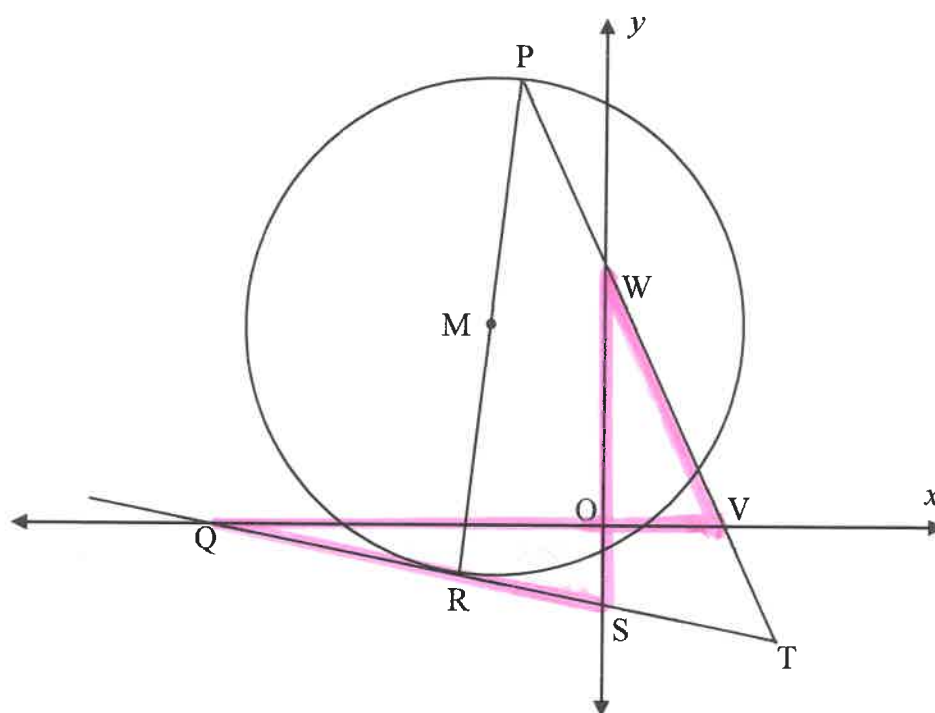
QUESTION 4



	Solution	Marks
4.1	$RE = \sqrt{(-4+2)^2 + (-1-2)^2} \checkmark$ $= \sqrt{13} \checkmark \text{ units}$	(2)
4.2	$M_{DR} = \left(\frac{4-4}{2}, \frac{3-1}{2} \right) = (0; 1) \checkmark$ $M_{AE} = \left(\frac{-2+2}{2}, \frac{2+0}{2} \right) = (0; 1) \checkmark$ $\therefore M_{DR} = M_{AE}$ $\therefore \text{DARE is a parm (Diags bisect)} \checkmark$	(3)

	Solution	Marks
4.3	$m_{DA} = \frac{3-0}{4-2} = \frac{3}{2} \quad \checkmark m_{DA}$ $y-0 = \frac{3}{2}(x-2) \quad \checkmark \text{ Can also use } y=mx+c$ $y = \frac{3}{2}x - 3 \quad \checkmark$	(3)
4.4	$m_{RE} = m_{DA}$ $\tan \theta = \frac{3}{2} \quad \checkmark m$ $\therefore \theta \approx 56^\circ \quad \checkmark \text{ rounding!}$	(2)
		[10]

QUESTION 5



	Solution	Marks
5.1	$M(-3; 4)$	(2)
5.2	$\sqrt{26} \approx 5,10$ units ✓ Accept either	(1)
5.3	$m_{PR} = 5$ ✓ (rad $\perp$ tan) $y - 4 = 5(x + 3)$ ✓ Can use $y = 5x + 19$ ✓ $y = mx + c$	(3)

	Solution	Marks
5.4	$(x+3)^2 + (y-4)^2 = 26 \quad y = 5x + 19$ $(x+3)^2 + (5x+19-4)^2 = 26 \quad \checkmark m$ $(x+3)^2 + (5x+15)^2 = 26$ $(x+3)^2 + (5(x+3))^2 = 26$ $(x+3)^2 + 25(x+3)^2 = 26 \quad \checkmark$ $26(x+3)^2 = 26$ $(x+3)^2 = 1 \quad \checkmark$ $x+3 = 1 \quad \text{or} \quad x+3 = -1$ $x = -2 \quad \checkmark \text{ x-values} \quad x = -4$ $N/A \quad \therefore y = -1$ $R(-4; -1)$ Must be in coordinates	(6)
5.5	$-1 = -\frac{1}{5}(-4) + k \quad \checkmark \text{subs } R$ $\therefore k = -\frac{9}{5} \quad \checkmark$	(2)
5.6	$\tan \hat{VQS} = -\frac{1}{5} \quad \checkmark m$ $\hat{VQS} = 11.31^\circ \quad \checkmark$ $\hat{VQS} = \hat{OWV} \quad \checkmark s$ $\therefore WVSQ \text{ is a cyclic quad } \checkmark R$ (converse $\angle$'s in the same segment) (chord subtends = $\angle$'s)	(4)
		[18]

5.4 Alt.

$$(x+3)^2 + (5x+19-4)^2 = 26 \quad \checkmark m$$

$$(x+3)^2 + (5x+15)^2 = 26$$

$$x^2 + 6x + 9 + 25x^2 + 150x + 225 = 26 \quad \checkmark$$

$$26x^2 + 156x + 208 = 0$$

$$x^2 + 6x + 8 = 0 \quad \checkmark$$

$$(x+4)(x+2) = 0$$

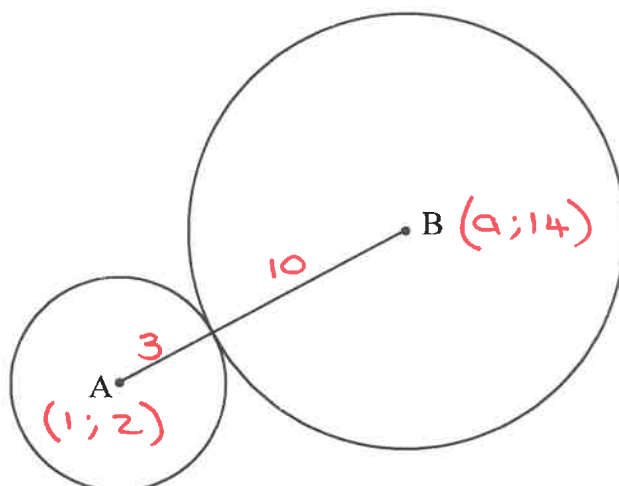
$$x = -4 \quad \checkmark \text{ or } \overset{x\text{-values}}{x = -2} \quad \text{N/A}$$

$$\therefore y = -1$$

$$\therefore R(\overset{\checkmark}{-4}; \overset{\checkmark}{-1})$$

(6)

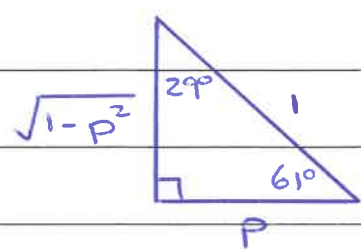
QUESTION 6



	Solution	Marks
6.1	$x^2 - 2x + (-1)^2 + y^2 - 4y + (-2)^2 = 4 + 1 + 4$ $(x-1)^2 + (y-2)^2 = 9 \quad \checkmark \text{SF}$ $\therefore A(1; 2) \quad \checkmark A$ $r = 3 \text{ units} \quad \checkmark r$	$\checkmark m$ (4)
6.2	$AB = 10 + 3 = 13$ $\sqrt{(a-1)^2 + (14-2)^2} = 13 \quad \checkmark AB=13$ $a^2 - 2a + 1 + 144 = 169 \quad \checkmark ()^2 = ()^2$ $a^2 - 2a - 24 = 0 \quad \checkmark \text{SF}$ $(a-6)(a+4) = 0$ $\underline{a=6} \quad \checkmark \quad \text{or} \quad a=-4 \quad \checkmark \text{reject}$ N/A	 (5)
		[9]

QUESTION 7

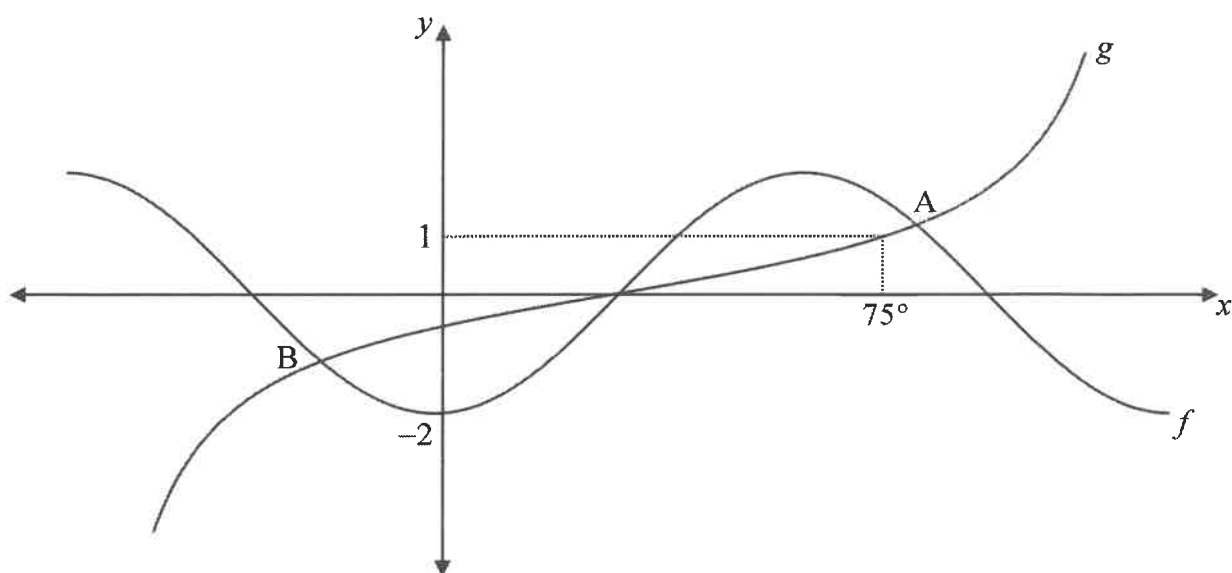
	Solution	Marks
7.1.1	$\cos(70+\theta)\cos(10+\theta)+\sin(70+\theta)\sin(10+\theta)$ $= \cos(70+\theta-(10+\theta)) \quad \checkmark \text{ compound}$ $= \cos 60^\circ \quad \checkmark$ $= \frac{1}{2} \quad \checkmark$	(3)
7.1.2	$\frac{-2\sin\theta\cos\theta}{\cos^2(180-\theta)+\cos(90-\theta)\sin(\theta-180)}$ $= \frac{-\sin 2\theta \quad \checkmark}{\cos^2\theta + \sin\theta \cdot (-\sin\theta) \quad \checkmark}$ $= \frac{-\sin 2\theta}{\cos^2\theta - \sin^2\theta}$ $= \frac{-\sin 2\theta}{\cos 2\theta} \quad \checkmark$ $= -\tan 2\theta \quad \checkmark$	(6)

	Solution	Marks		
7.2.1	$\tan(-42^\circ)$ $= -\tan 61^\circ \checkmark$ $= -\frac{\sqrt{1-p^2}}{p} \checkmark$ 	(3)		
7.2.2	<table><tr><td>$\sin 122^\circ$ $= \sin(2(61^\circ)) \checkmark$ $= 2\sin 61^\circ \cos 61^\circ \checkmark$ $= 2\sqrt{1-p^2} \cdot p \checkmark$ $= 2p\sqrt{1-p^2}$</td><td>Alt: $\sin 122^\circ$ $= \sin 58^\circ \checkmark$ $= 2\sin 29^\circ \cos 29^\circ \checkmark$ $= 2p\sqrt{1-p^2} \checkmark$</td></tr></table>	$\sin 122^\circ$ $= \sin(2(61^\circ)) \checkmark$ $= 2\sin 61^\circ \cos 61^\circ \checkmark$ $= 2\sqrt{1-p^2} \cdot p \checkmark$ $= 2p\sqrt{1-p^2}$	Alt: $\sin 122^\circ$ $= \sin 58^\circ \checkmark$ $= 2\sin 29^\circ \cos 29^\circ \checkmark$ $= 2p\sqrt{1-p^2} \checkmark$	(3)
$\sin 122^\circ$ $= \sin(2(61^\circ)) \checkmark$ $= 2\sin 61^\circ \cos 61^\circ \checkmark$ $= 2\sqrt{1-p^2} \cdot p \checkmark$ $= 2p\sqrt{1-p^2}$	Alt: $\sin 122^\circ$ $= \sin 58^\circ \checkmark$ $= 2\sin 29^\circ \cos 29^\circ \checkmark$ $= 2p\sqrt{1-p^2} \checkmark$			
7.3	$\cos(M+N) = \cos(M - (-N)) \checkmark$ $= \cos M \cos(-N) + \sin M \sin(-N) \checkmark$ $= \cos M \cos N + \sin M (-\sin N) \checkmark$ $= \cos M \cos N - \sin M \sin N$	(3)		

	Solution	Marks
7.4.1	$\frac{3\cos 2x + 3\cos^2 x + 9\sin^2 x}{4 - 4\sin^2 x} = \frac{3}{2\cos^2 x}$ $\text{LHS} = \frac{3(\cos^2 x - \sin^2 x) + 3\cos^2 x + 9\sin^2 x}{4(1 - \sin^2 x)}$ $= \frac{6\cos^2 x + 6\sin^2 x}{4\cos^2 x}$ $= \frac{6(\cos^2 x + \sin^2 x)}{4\cos^2 x}$ $= \frac{6(1)}{4\cos^2 x}$ $= \frac{3}{2\cos^2 x} = \text{RHS}$ $\therefore \text{LHS} = \text{RHS}$	(5)
7.4.2	$\cos^2 x = 0$ $x = 90^\circ + k \cdot 180^\circ, k \in \mathbb{Z}$ $\therefore x \in \{90^\circ\}$	(2)

	Solution	Marks
7.5	$\sin x \cos 20^\circ + \cos x \sin 20^\circ = \sin 3x$ $\sin(x + 20^\circ) = \sin 3x$ ref L = $3x$ ✓m I: $x + 20^\circ = 3x + k \cdot 360^\circ$ $-2x = -20^\circ + k \cdot 360^\circ$ $x = 10^\circ - k \cdot 180^\circ; k \in \mathbb{Z}$ ✓ or ✓ Quadrants II: $x + 20^\circ = 180^\circ - 3x + k \cdot 360^\circ$ $4x = 160^\circ + k \cdot 360^\circ$ $x = 40^\circ + k \cdot 90^\circ; k \in \mathbb{Z}$ ✓ 	(5)
		[30]

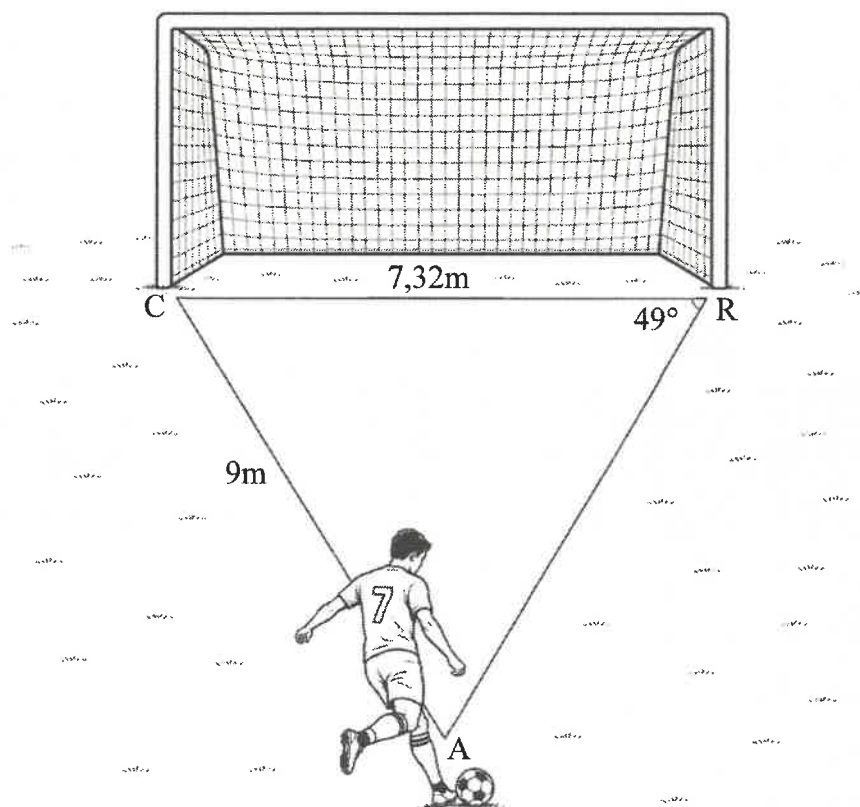
QUESTION 8



	Solution	Marks
8.1	$a = -2$ ✓	(3)
	$b = 3$ ✓	
	$c = 30^\circ$ ✓	
8.2	120° ✓	(1)
8.3	$x = 120^\circ$ or $x = -60^\circ$ ✓ Accept 1	(1)
8.4	$B(-18,5^\circ; -1,13)$	(2)
	✓ ✓	
8.5.1	$x = 60^\circ$ ✓	(1)
8.5.2	$-30^\circ \leq x \leq 90^\circ$ ✓	(1)
		[9]

QUESTION 9

9.1

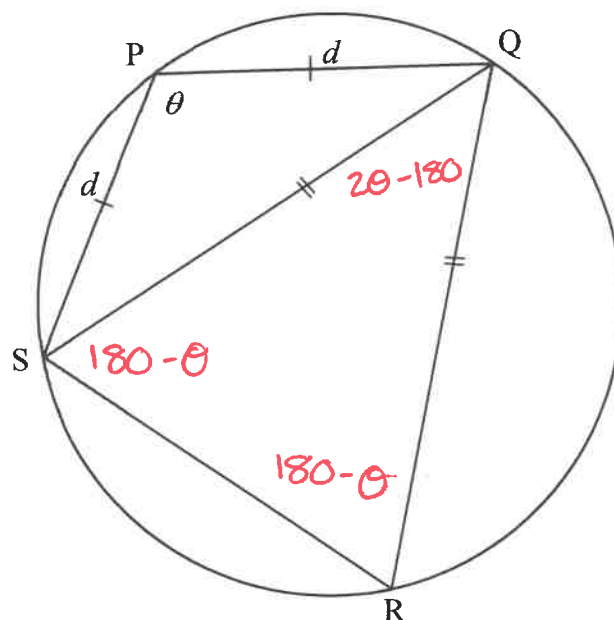


	Solution	Marks
9.1	$\frac{\sin \hat{A}}{CR} = \frac{\sin \hat{R}}{CA} \quad \checkmark \text{Use sin rule}$	
	$\frac{\sin \hat{A}}{7,32} = \frac{\sin 49^\circ}{9} \quad \checkmark \text{Subs}$	
	$\sin \hat{A} = \frac{7,32 \cdot \sin 49^\circ}{9}$	
	$\sin \hat{A} = 0,6138...$	
	$\therefore \hat{A} \approx 38^\circ \quad \checkmark$	

(3)

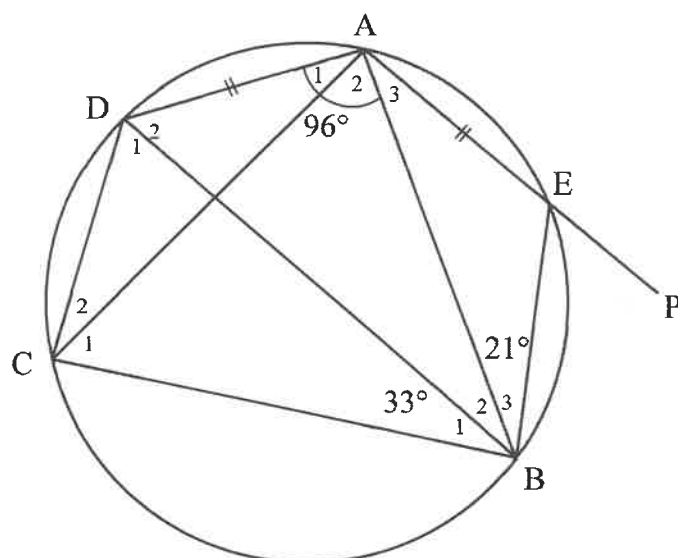
	Solution	Marks
9.2.2	$\hat{R} = 180^\circ - \theta$ (Opp $\angle$ s of cyclic quad) $\therefore \hat{S} = 180^\circ - \theta$ ($\angle$ s opp = sides) $\therefore \hat{SQR} = 2\theta - 180^\circ \checkmark^S$ ($\angle$ s in a Δ) $\text{Area } \Delta QRS = \frac{1}{2} SQ \cdot QR \cdot \sin \hat{Q} \checkmark^{\text{Area rule}}$ $= \frac{1}{2} \left(d\sqrt{2(1-\cos\theta)} \right)^2 \cdot \sin(2\theta - 180^\circ) \checkmark^{\text{subs}}$ $= \frac{1}{2} d^2 (2(1-\cos\theta)) \cdot (-\sin 2\theta) \checkmark^{\text{Reduction}}$ $= -d^2 (1-\cos\theta) \sin 2\theta$	(4)
		[10]

9.2



	Solution	Marks
9.2.1	In $\triangle PQS$: $QS^2 = PS^2 + PQ^2 - 2PS \cdot PQ \cdot \cos \hat{P} \quad \checkmark \text{ use cos rule}$ $QS^2 = d^2 + d^2 - 2d \cdot d \cdot \cos \theta \quad \checkmark \text{ subs}$ $QS^2 = 2d^2 - 2d^2 \cos \theta$ $QS^2 = d^2 (2 - 2\cos \theta) \quad \checkmark \text{ Factors}$ $QS = d \sqrt{2 - 2\cos \theta}$	(3)

QUESTION 10



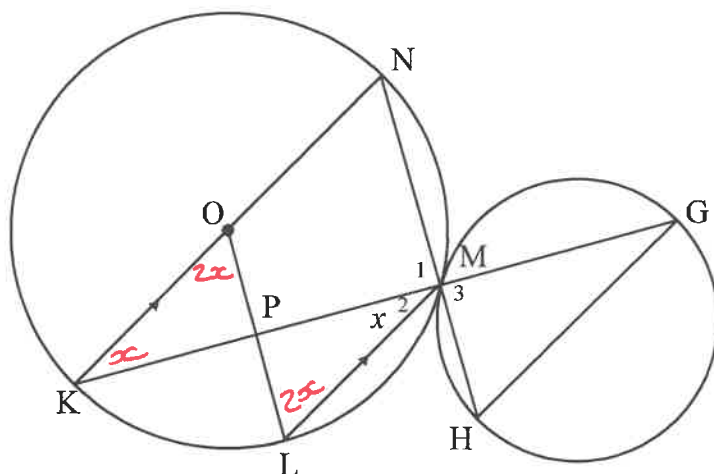
	Solution	Marks
10.1	$\hat{B} = 21^\circ \text{ S/R } (= \text{chords}; = \angle\text{'s})$	(2)
	$\hat{C}_2 = 21^\circ \text{ S/R } (= \text{chords}; = \angle\text{'s})$	
10.2	$\hat{C}_1 + \hat{C}_2 + \hat{DAB} = 180^\circ \text{ (opp } \angle\text{'s of cyclic quad)}$	(3)
	$\therefore \hat{C}_1 = 63^\circ \text{ ✓ A}$	
10.3	$\hat{BEP} = \hat{C}_1 \text{ (Ext } \angle \text{ of cyclic quad)}$	(2)
	$\therefore \hat{BEP} = 63^\circ \text{ ✓ A}$	
10.4	$\hat{A}_1 = \hat{B}_1 = 33^\circ \text{ S/R } (\angle\text{'s in the same segment})$	(3)
	$\therefore \hat{A}_2 = 63^\circ \text{ ✓}$	
	$\therefore \hat{A}_2 = \hat{C}_1$	
	$\therefore CB = AB \text{ (Sides opp } \angle\text{'s)}$	
		[10]

QUESTION 10

	Solution	Marks
10.1	In $\triangle POR$: $\hat{R}_1 = \hat{P}$ $\angle$s opp radii ✓ $\hat{O}_1 = \hat{R}_1 + \hat{P}$ Ext $\angle$ of $\triangle$ ✓ $\therefore \hat{O}_1 = 2\hat{R}_1$ ✓ Similarly, $\hat{O}_2 = 2\hat{R}_2$ $\therefore \hat{POQ} = \hat{O}_2 - \hat{O}_1$ $= 2\hat{R}_1 - 2\hat{R}_2$ $= 2(\hat{R}_1 - \hat{R}_2)$ $= 2\hat{P}RQ$ ✓	

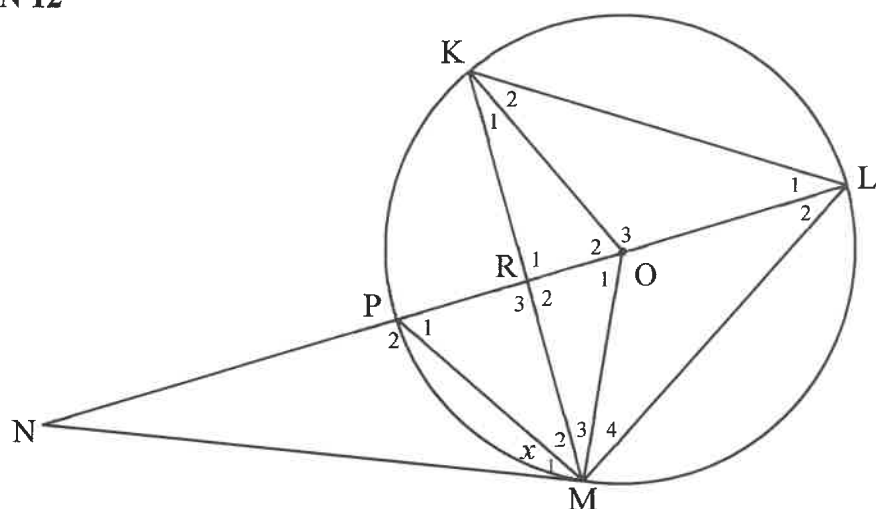
(5)

11.2



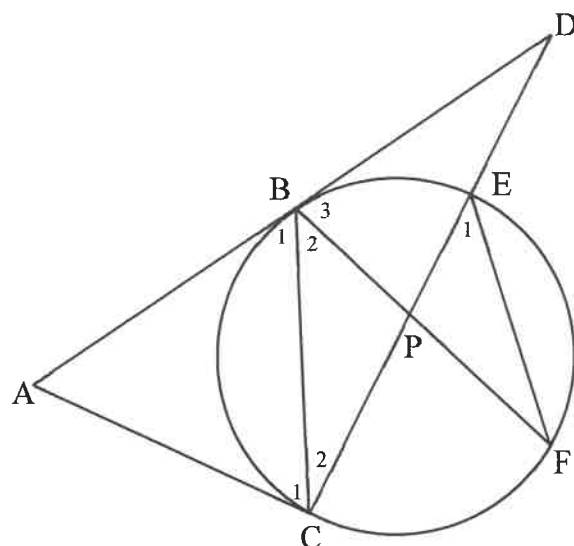
	Solution	Marks
11.2.1	$\hat{KOL} = 2x$ ✓ ^S (∠ at centre = 2 × ∠ at circ) ✓ ^R $\hat{L} = \hat{KOL}$ (Alt ∠s ; $OK \parallel LM$) $\therefore \hat{L} = 2x$ ✓ ^S ✓ ^R	(4)
11.2.2	$\hat{M}_1 = 90^\circ$ ✓ ^S ✓ ^R (∠ in semi ∅) $\therefore \hat{M}_3 = 90^\circ$ ✓ ^S ✓ ^R (Vert opp ∠s) $\therefore GH$ is diameter (converse ∠ in semi ∅) ✓ ^R (chord subtends 90°)	(3)
11.2.3	$\hat{MPL}$ must be 90° ✓ ^S ✓ ^R (line from centre to midpoint chord) $\therefore 3x + 90 = 180^\circ$ ✓ ^S ✓ ^R (∠s in a ∆) $3x = 90$ $x = 30^\circ$ ✓ ^A	(3)
		[15]

QUESTION 12



	Solution	Marks
12.1	$\hat{O}MN = 90^\circ$ ✓S (rad $\perp$ tan) $\hat{P}ML = 90^\circ$ ✓S ($\angle$ in semi $\odot$)	(2)
12.2 (a)	$\hat{L}_2 = x$ ✓S (Tan Chord Thrm) ✓R	(2)
12.2 (b)	$\hat{M}_4 = x$ ✓S (L's opp radii) ✓R	(2)
12.2 (c)	$\hat{N} + \hat{N}ML + \hat{L}_2 = 180^\circ$ (L's in a Δ) $\hat{N} = 180 - x - (90 + x)$ ✓ $\hat{N} = 90 - 2x$ ✓	(3)
12.3	$\hat{N}KO = 90^\circ$ ✓S/R (rad $\perp$ tan) $\therefore \hat{N}KO + \hat{O}MN = 180^\circ$ $\therefore$ KOMN is cyclic quad ✓R (converse opp $\angle$ s cyclic quad)	(2)
	(opp $\angle$ s of quad supp)	[11]

QUESTION 13



	Solution	Marks
13.1	$AB = AC$ $\checkmark_R$ (Tangents from same point) $\therefore \hat{C}_1 = \hat{B}_1$ $\checkmark_R$ (L's opp = sides) $\hat{B}_1 = \hat{D} + \hat{C}_2$ $\checkmark_R$ (Ext L of Δ) $\hat{C}_2 = \hat{F}$ $\checkmark_R$ (L's in same segment) $\therefore \hat{C}_1 = \hat{D} + \hat{F}$	(4)
13.2	$\hat{E}_1 = \hat{D}$ (Given) $\hat{E}_1 = \hat{B}_2$ $\checkmark_R$ (L's in same segment) $\therefore \hat{B}_2 = \hat{D}$ $\checkmark_S$ BC is a tangent (converse tan chord Thrm) $\checkmark_R$	(3)
		[7]